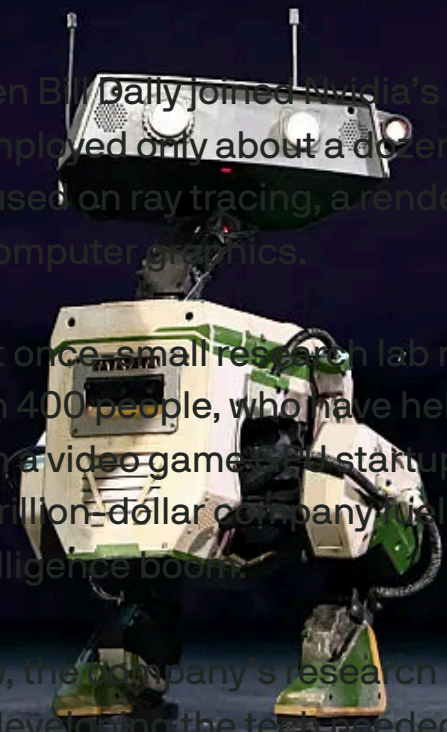


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ROBOTICS



How a once-tiny
research lab helped
Nvidia become a \$4
trillion-dollar company



When Bill Dally joined Nvidia's research lab in 2009, it employed only about a dozen people and was focused on ray tracing, a rendering technique used in computer graphics.

That once-small research lab now employs more than 400 people, who have helped transform Nvidia from a video game and startup in the nineties to a \$4 trillion-dollar company fueling the artificial intelligence boom.

Now, the company's research lab has its sights set on developing the tech needed to power robotics and AI. And some of that lab work is already showing up in products. The company unveiled Monday a [new set of world AI models](#), libraries, and other infrastructure for robotics developers.

Dally, now Nvidia's chief scientist, started consulting for Nvidia in 2003 while he was working at Stanford. When he was ready to step down from being the department chair of Stanford's computer science department a few years later, he planned to take a sabbatical. Nvidia had a different idea.

David Kirk, who was running the research lab at the time, and Nvidia CEO Jensen Huang, thought a more permanent position at the research lab was a better idea. Dally told TechCrunch the pair put on a "full-court press" on why he should join Nvidia's research lab and eventually convinced him.

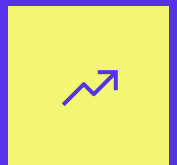
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BILL DALLY

IMAGE CREDITS:NVIDIA

“It would up being kind of a perfect

t fit for my interests and my talents,” Dally said. “I think everybody’s always searching for the place in life where they can make the biggest contribution to the world. And I think for me, it’s definitely Nvidia.”

When Dally took over the lab in 2009, expansion was first and foremost. Researchers started working on areas outside of ray tracing right away, including circuit design and VLSI, or very large-scale integration, a process that combines millions of transistors on a single chip.

The research lab hasn’t stopped expanding since.

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“We try to figure out what will make the most positive difference for the company because we’re constantly seeing exciting new areas, but some of them, they do great work, but we have trouble saying if [we’ll be] wildly successful at this,” Dally said.

For a while that was building better GPUs for artificial intelligence. Nvidia was early to the future AI boom and started tinkering with the idea of AI GPUs in 2010 — more than a decade before the current AI frenzy.

“We said this is amazing, this is gonna completely change the world,” Dally said. “We have to start doubling down on this and Jensen believed that when I told him that. We started specializing our GPUs for it and developing lots of software to support it, engaging with the researchers all around the world who were doing it, long before it was clearly relevant.”

Physical AI focus

Now, as Nvidia holds a commanding lead in the AI GPU market, the tech company has started to seek out new areas of demand beyond AI data centers. That search has led Nvidia to physical AI and robotics.

“I think eventually robots are going to be a huge player in the world and we want to basically be making the brains of all the robots,” Dally said. “To do that we need to start developing the key technologies.”

That’s where Sanja Fidler, the vice president of AI research at Nvidia, comes in. Fidler joined Nvidia’s research lab in 2018. At the time, she was already working on simulation models for robots with a team of students at MIT. When she told Huang about what they were working on at a researchers’ reception, he was interested.

“I could not resist joining,” Fidler told TechCrunch in an interview. “It’s just such a great topic fit and at the same time was also such a great culture fit. Jensen told me, come work with me, not with us, not for us.”

She joined Nvidia and got to work creating a research lab in Toronto called Omniverse, an Nvidia platform, that was focused on building simulations for physical AI.

The first challenge to building these simulated worlds was finding the necessary 3D data, Fidler said. This included finding the proper volume of potential images to use and building the technology needed to turn these images into 3D renditions the simulators could use.

“We invested in this technology called differentiable rendering, which essentially makes rendering amenable to AI,” Fidler said. “You go [from] rendering means from



SANJA FIDLER

IMAGE CREDITS: NVIDIA

3D to image or video. And we want it to go the other way.”

World models

Omniverse released the first version of its model that turns images into 3D models, [GANverse3D](#), in 2021. Then it got to work on figuring out the same process for video. Fidler said they used videos from robots and self-driving cars to create these 3D models and simulations through its [Neural Reconstruction Engine](#), which the company first announced in 2022.

She added these technologies were the backbone of the company’s [Cosmos family of world AI models](#) that were announced at CES in January.

Now, the lab is focused on making these models faster. When you play a video game or simulation you want the tech to be able to respond in real time, Fidler said, for robots they are working to make the reaction time even faster.

“The robot doesn’t need to watch the world in the same time, in the same way as the world works,” Fidler said. “It can watch it like 100x faster. So if we can make this model significantly faster than they are today, they’re going to be tremendously useful for robotic or physical AI applications.”

The company continues to make progress on this

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“We’re making huge progress and I think AI has really been the enabler here,” Dally said. “Starting with visual AI for the robot perception, and then generative AI, that’s being hugely valuable for task and motion planning and manipulation. As we solve each of these individual little problems and as the amount of data we have to train our networks grows, these robots are going to grow.”

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Becca is a senior writer at TechCrunch that covers venture capital trends and startups. She previously covered the...

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HARDWARE



Nvidia said to be developing new, more powerful AI chip for sale in China

Ram Iyer — 5:58 AM PDT · August 19, 2025

IMAGE CREDITS:

AKIO KON/BLOOMBERG / GETTY IMAGES

The world's most valuable chipmaker is far from giving up its desire to retain China as a key growth market. Nvidia is apparently putting together a new AI chip meant for sale in China that's half as powerful as its flagship B300 Blackwell GPU, Reuters [reported](#), citing anonymous sources.

This new chip, codenamed B30A, would be more powerful than the H20 GPUs the company is currently allowed to sell to China, the report said, though its design will be single-die in nature, unlike the dual-die design used in the more powerful B300 GPUs. The B30A would still have similar features to the H20, like fast data transmission, support for NVLink, and high-bandwidth memory.

The development of the B30A seems to be separate from [another chip](#) that the company is said to be developing for sale in the country, Reuters reported.

"We evaluate a variety of products for our roadmap, so that we can be prepared to compete to the extent that governments allow. Everything we offer is with the full approval of the applicable authorities and designed solely for beneficial commercial use," Nvidia said in an emailed statement.

The news comes as the Trump administration in recent weeks has [relaxed its stance](#) on letting chipmakers export high-performance AI chips to China, though Reuters cited its sources as saying that approvals for this new chip aren't certain at all.

As geopolitical tensions between Beijing and Washington are increasingly affected by the race to develop AI, critics [argue](#) that the U.S. must maintain its lead by controlling the supply of necessary technology to China. But Nvidia and its peers — finding themselves [selling the equivalent of shovels during a gold rush](#) — maintain that ceding the valuable market of China to rivals like Huawei would be tantamount to giving up entirely.

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Ram is a financial and tech reporter and editor. He covered North American and European M&A, equity, regulatory...

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ENTERPRISE



SoftBank makes \$2B investment in Intel

Kirsten Korosec — 5:23 PM PDT · August 18, 2025

Japanese conglomerate SoftBank has agreed to make a \$2 billion investment in Intel in a deal described as a commitment to advanced technology and semiconductors in the United States.

The agreement, in which SoftBank will buy Intel common stock, [was announced Monday](#) after markets closed. SoftBank will pay \$23 per share of Intel common stock. Shares of Intel, which closed at \$23.66, popped more than 5% in after-hours trading.

SoftBank Group Chairman and CEO Masayoshi Son said in a statement that the “strategic investment reflects our belief that advanced semiconductor manufacturing and supply will further expand in the United States, with Intel playing a critical role.”

The investment provides validation for Intel, which has been overshadowed in recent years by competitors like Nvidia. It also reflects SoftBank’s renewed interest in the U.S., particularly around AI chips. SoftBank recently [purchased a factory](#) in Lordstown, Ohio, owned by Foxconn, as part of a plan to build AI data centers.

Intel, steered by new CEO Lip-Bu Tan, is in the midst of a restructuring that aims to streamline the semiconductor business and focus on its core client and data center portfolio. Earlier this summer, Intel [shuttered its automotive architecture business](#) and laid off most of its staff. It also announced plans to reduce its Intel Foundry division workforce [between 15% and 20%](#).

Tan has also had to navigate political landmines in recent weeks as President Donald Trump [called for his resignation](#) due to conflicts of interest — an accusation that was made without evidence — and his administration reportedly had discussions to [take a stake in Intel](#).

The SoftBank-Intel deal comes just days after the Trump administration threatened [new tariffs](#) on imported semiconductor chips as part of his strategy to boost domestic production.

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